SEAT NO. CS-23007

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR (Computer Systems Engineering)

SPRING SEMESTER EXAMINATIONS 2026

Time : 3 Hours | **Batch 2023** | **Dated : 01-JUN-26**

Max Marks : 60

Engineering Economics & Management - EF-305

Instructions: Attempt all the questions.

Q1. Hi-Tech, is a sole-proprietorship software company operated by Hi-Tech owner Mr. Jack. On July 1, 2024, Hi-Tech, has the following assets and liabilities: cash, $1,000; accounts receivable, $3,200; supplies, $850; land, $10,000; accounts payable, $1,530. Office space and office equipment are currently being rented, pending the construction of an office complex on land purchased last year. Business transactions during July-2024 are summarized as follows: **(CLO 1)**

- Received cash from clients for services, $3,928.
- Paid creditors on account, $1,055.
- Received cash from Cecil Jameson as an additional investment, $3,700.
- Paid office rent for the month, $1,200.
- Charged clients for legal services on account, $2,025.
- Purchased supplies on account, $245.
- Received cash from clients on account, $3,000.
- Received invoice for License expense from government for July (to be paid on August 10), $1,635.
- Paid the following: wages expense, $850; answering service expense, $250; utilities expense, $325; and miscellaneous expense, $75.
- Determined that the cost of supplies on hand was $980; therefore, the cost of supplies used during the month was $115.
- Jameson withdrew $1,000 in cash from the business for personal use.

a. **Record** the amount of assets, liabilities, and owner's equity (Mr. Jack's capital) as of July 1, 2016 (as opening balance of the month of July) in Basic Accounting equation form (i.e. Assets = Liabilities + Owner's Capital). **(03)**

b. Use the above equation with opening balances and **record** each of the above transaction in sequence to show the effect on the two sides of basic accounting equation to maintain a balance of equation and **match** the closing balance on both sides of equation. **(07)**

Q2. You have been given the following data from the "Novel Tech Company" for the year ending December 31, 2024. **Tabulate** an income statement and a balance sheet from the data. **(CLO 1) (04+06)**

Accounts Payable	$27500	Depreciation Expense, buildings	900
Accounts Receivable	32000	Government bonds	25000
Advertising Expense	2500	Income Taxes	9350
Bad Debt Expense	1100	Insurance Expense	1100
Building, Net	14000	Inventory, December 31, 2024	42000
Cash	45250	Land	25000
Common Stock	125000	Machinery, net	3400
Cost of Goods Sold	311250	Mortgage due May 30, 2027	5000
Office Equipment, net	5250	Prepaid expenses	3000
Office Supplies Expense	2025	Retained Earnings	?
Other Expenses	7850	Sales	421400
Salaries Expense	69025	Tax payable	2500
Total Expenses	84500	Wage payable	600

TR = TC   TC =

=

Q3. a. A three-year-old extruder used in making plastic yogurt cups has a current book value of $168 750. The declining-balance method of depreciation with a rate d = 0.25 is used to determine depreciation charges. **Compute** what was its original price and what will its book value be two years from now.
(CLO 1) (03)

b. An asset costs $14 000. **Compute** what declining-balance depreciation rate would result in the scrap value of $3000 after seven years.
(CLO 1) (02)

Q4. Consider public policy aimed at controlling petrol consumption. **(CLO 2)**

a. Studies indicate that the price elasticity of demand for petrol is about 0.4. If a liter of petrol is currently priced at Rs. 6 and the government wants to reduce petrol consumption by 20 per cent, **compute** by how much should it increase the price. (-3) **(03)**

b. If the government permanently increases the price of petrol, **discuss** how will it affect the price elasticity of demand for petrol in long run? (-2) **(02)**

Q5. A firm has the following information available to its managers: Fixed costs (FC) are Rs. 1500, total variable costs (VC) are Rs. 2000 and firms' total revenue (TR) is Rs 5000. **Compute** the following:
(CLO 2) (05)

a. Profit-Volume Ratio
b. Break-even sales in rupees ×
c. Margin of safety in rupees

(-4)

Q6. **Calculate** the followings: **(CLO 3)**

a. Greg wants to have $50 000. He will invest $20 000 today in investment certificates that pay 8 percent nominal interest, compounded quarterly. How long will it take him to reach his goal?
(03)

b. If an investment offers 25 percent nominal interest compounded monthly. Compute the effective annual interest rate. (-4) **(04)**

c. Fred wants to save up for an automobile. What amount must he put in his bank account each month to save $10 000 in two years if the bank pays 6 percent interest compounded monthly?
(-4) **(04)**

d. A car loan requires 30 monthly payments of $199.00, starting *today*. At an annual rate of 12 percent compounded monthly, how much money is being lent? **(04)**

e. Estimated revenues of a local company are $80,000 for first year. Expected to rise by uniform amount to a total of $200,000 by the end of year 9. Determine the Gradient. **(04)**

Q7. Investment proposals A and B have the net cash flows as follows: **(CLO 3) (06)**

Proposals	End of Years				
	0	1	2	3	4
A (Rs.)	-10000	3000	3000	7000	6000
B (Rs.)	-10000	6000	6000	3000	3000

If rate of return is i = 18%, **Apply** present worth method to decide which proposal should be selected.

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NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR(Computer Systems Engineering)

SPRING SEMESTER EXAMINATIONS 2026

Time : 3 Hours | **Batch 2023** | **Dated : 03-JUN-26**

Max Marks : 60

Microprocessor Based System Design - CS-301

Instructions:

1. Attempt all questions
2. Show all calculations and intermediate results
3. State your assumptions where necessary

Q1 – CLO-1 (Application – C3) – PLO-1 (Engineering Knowledge)

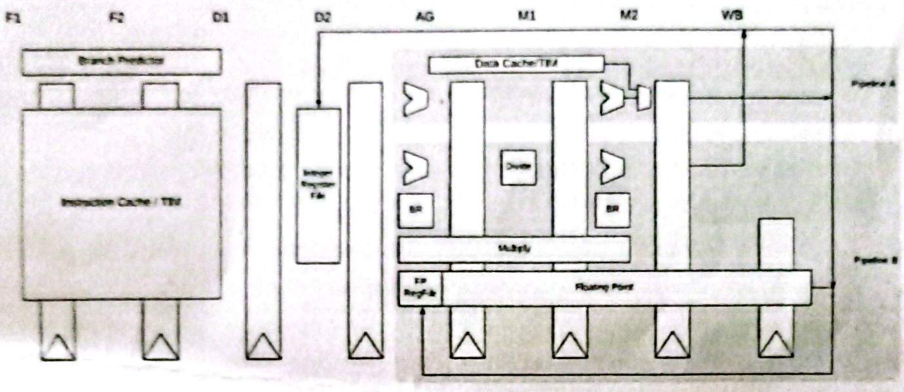


Fig. 1 Pipeline stages of SiFive U74 processor

Q1 a Show through a pipeline execution diagram how will the following code run on the dual-issue U74 processor (Fig. 1) assuming all instruction and data accesses are hit in the respective caches. Show the first 10 cycles of execution only. Consider an 8-entry BTB that stores the predicted target addresses for taken branches only. The BTB already has an entry for the unconditional branch at address 0x501C. **[04]**

```
0x5000      again:      beq    x11, x0, exit      // x11 = 50 currently
0x5004                  ld     x5, 100(x10)
0x5008                  ld     x6, 108(x10)
0x500C                  add    x5, x5, x6
0x5010                  sd     x5, 100(x10)
0x5014                  addi   x11, x11, -1
0x5018                  addi   x10, x10, 8
0x501C                  jal    x0, again
0x5020            exit: addi   x10, x0, 1
0x5024                  jalr   x0, x1, 0
```

Q1 b When the code finally finishes, show what new entry will there be in the BTB. **[01]**

Q1 c If there are two entries in the RAS where the top of stack has value 0x3200 and the other entry is 0x1200 which was stored earlier, what will happen when instruction at address 0x5024 will execute? **[01]**

Q2 – CLO-1 (Application – C3) – PLO-1 (Engineering Knowledge)

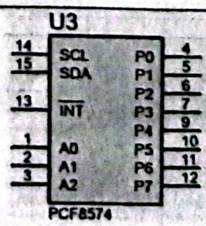
Q2 a The memory system uses a SECDED Hamming code for protecting 16-bit data words. Determine the minimum number of Hamming parity bits required and their positions. **[01]**

Q2 b	Construct the complete code word for this memory system by calculating all the parity bits if the data to be stored is D = 1011 0110 1101 0011	02
Q2 c	Suppose the received codeword for the same parity bits as calculated in part b is D = 1011 0111 1101 0011 Using syndrome calculation determine the syndrome bits and recover the original data word.	02
Q2 d	Now suppose two bits are flipped simultaneously, how will SECDED distinguish this from a single-bit error?	01

Q3 – CLO-1 (Application – C3) – PLO-1 (Engineering Knowledge)

Q3 a	An embedded processor uses dynamic voltage and frequency scaling and can be operated at three different voltage levels, 2.0 V, 3.0 V and 4.0 V. The maximum operating frequency is 100 MHz at 4 V. When the voltages are scaled the max frequency is also reduced linearly. Average energy consumption per cycle is 50 nJ at 2.0 V. Solve the following questions assuming a periodic task is run every 3 minutes on this processor, which needs 15 billion cycles for completion. i. Calculate the energy consumption per cycle, frequency and cycle time at all remaining voltage levels. ii. Schedule the task such that performance is maximized. What is the energy consumption of this schedule? iii. Schedule the task such that energy consumption is minimized, using DVFS. What is the reduction achieved in the energy consumption from the previous schedule?	06
Q3 b	Suppose a workload trace on a processor when dynamic power management is not applied is as follows: 40, 400, 200, 50, 60, 800, 340, 100, 1000 (ms) The underlined numbers refer to the duration of an active period. The processor can be in either of the two states RUN or the power-saving state SLEEP. We assume the processor starts in the RUN state. The transition time from RUN to SLEEP is 80 μs and is considered negligible as compared to 250 ms required from SLEEP to RUN. The power consumption during the RUN state and during transitions is 0.6 W. The power consumption during the sleep state is only 0.3 mW. Calculate total run time, and energy consumption of the workload for threshold-based predictive shutdown DPM policy with a threshold of 80 ms.	02

Q4 – CLO-2 (Analysis – C4) – PLO-2 (Problem Analysis)

Q4 a	Illustrate how to interface an external SRAM of 16 KB and Flash memory of 8 KB with external memory interface of ATMega162 [PA0-PA7/AD0-AD7, PC7-PC0/A15-A8, PE1/ALE, PD6/WR' and PD7/RD']. SRAM and Flash memory interface include appropriate number of address lines, D0-D7 data bus, OE', WE' and CS (Chip select). You may use latch 74x573. No embedded C code required.	[5]
Q4 b	Using Overflow Interrupt in Timer/Counter0, figure out toggling of an LED at every 2 seconds using normal mode of operation. Consider MCU clock = 16 MHz, LED connected to PB0, Timer0 pre-scaler = 1024 [CS02 CS01 CS00=101 in TCCR0), TOIE0 is in TIMSK register, Interrupt vector name: TIMER0_OVF_vect. Provide embedded C code. No interface diagram required.	[5]
Q4 c	Consider the Remote IO Expander IC PCF8574 that supports I²C based communication. Interface two such IC's with ATMega16 PC1/SDA and PC0/SCL lines, first to take input from 8-bit DIP switches and second to show some outputs on 8-bit LED bank. Set the address bits accordingly. Outline all connections neatly. No embedded C code required. 	[5]

Q5 – CLO-1 (Application – C3) – PLO-

Q5 a For the Timer/Counter0 in CTC mode, if the microcontroller clock frequency is 8 MHz generate a 2 KHz waveform at OC0 (PB0). Place an appropriate value in OCR0 and an appropriate pre-scaler value, [CS02 CS01 CS00 (Pre-scaler): 001(1), 010 (8), 011 (64), 100(256), 101(1024), WGM01 WGM00:10 (CTC). COM01 COM00: 01 (Toggle OC0 on compare match) in TCCR0]. Provide embedded C code. No interface diagram required. **[3]**

Q5 b Compute the value of TWBR (Two Wire Bit Rate Register) for SCL frequency of 1.5 KHz when CPU frequency is 8 MHz and TWPS pre-scaler value is 1. Also draw the timing diagram to demonstrate how I²C protocol works. **[3]**

Q5 c **[4]**

Bit	7	6	5	4	3	2	1	0	
	I	T	H	S	V	N	Z	C	SREG
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Demonstrate the use of Global Interrupt Enable (I) at bit 7 in status register SREG with a concrete example using USART protocol. [RXCIE0 in UCSR0B, Interrupt vector name: USART0_RXC_vect, pins PD1 for TXD0 and PD0 for RXD0. The Baud rate register value for FOSC=1 MHZ can be taken as 6. The initialization code is :

```
UCSR0B    |=    (1<<TXEN0)|(1<<RXEN0)|(1<<RXCIE0)    //Enable    receiver,
transmitter,  receive-complete IE
UCSR0C|=|(1<<<URSEL)|(3<<UCSZ00); // Set frame format: 8 bit data
```

No interface diagram required.

Q6 – CLO-2 (Analysis– C4 – PLO-1 (Problem Analysis)

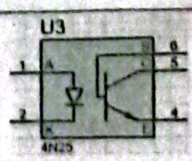
Q6 a Illustrate a circuit that takes 4-bit operand ***num*** from DIP switches through lower bits of port A and performs a comparison operation with the value $C)_{16}$. If num is greater than $C)_{16}$, it displays letter 'G' (as figure 6) on a common anode seven segment display. However, if it is lesser than $C)_{16}$, it displays 'L' and if it is equal then 'E' is displayed. Draw a neat interface block diagram and infer the embedded C code. **[4]**

Q6 b Explore the following user-defined functions to store user password to unlock a door in the on-chip EEPROM and authenticate it against password entered through keypad interfaced using keypad controller 74922 [inputs: rows and columns of keypad, outputs: DA (Data available) and DCBA (index of key pressed 0-15)]. Draw a neat Interface diagram and provide embedded C code. **[6]**

```
// EEPROM Write Function
void EEPROM_write(unsigned int
address, unsigned char data)
{
//Wait until previous write
//completes
      while (EECR & (1<<EEWE));
      // Set address and data
      EEAR = address;
      EEDR = data;
      // Enable write
      EECR |= (1<<EEMWE);
      // Start write
      EECR |= (1<<EEWE);}
```

```
// EEPROM Read Function
unsigned char
EEPROM_read(unsigned int address)
{// Wait until previous write
//completes
      while (EECR & (1<<EEWE));
      // Set address
      EEAR = address;
      // Start read
      EECR |= (1<<EERE);
      // Return data
      return EEDR;}
```

Q6 c Illustrate with a simple application, how an opto-coupler 4N25 as shown in figure, can be used to isolate a high voltage circuitry of 12 V from the microcontroller based circuit of 5V. Provide an appropriate embedded C code for your application. **[5]**



SEAT NO. ____________

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR(Computer Systems Engineering)

SPRING SEMESTER EXAMINATIONS 2026

Time : 3 Hours | Batch 2023 | Dated : 08-JUN-26

Max Marks : 60

Operating Systems - CS-329

INSTRUCTIONS:

1. Attempt all questions.
2. Make logical assumptions where necessary.
3. Questions may be attempted in any order, but all questions of a section and all parts of a question must be answered contiguously.

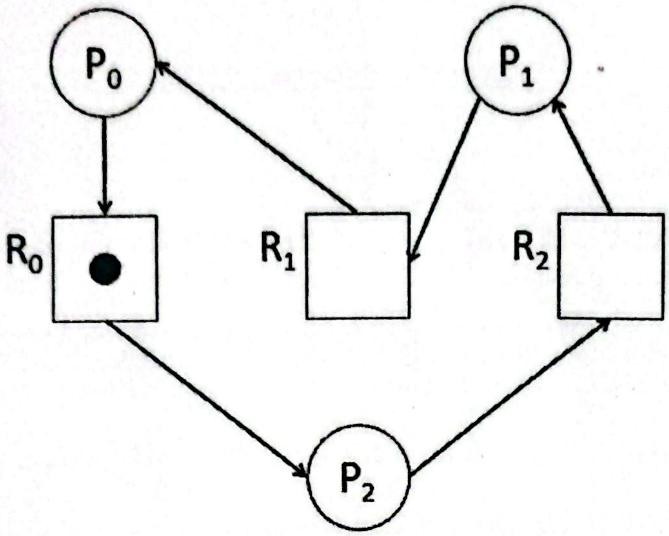
		CLO-1: Describe the fundamentals of operating system architecture (C2, PLO-1)	
1.	a)	**Discuss:** The SMP OS should provide graceful degradation in the face of processor failure.	2
	b)	**Explain** the virtual machine approach for multicore OS design, and how it can be advantageous.	2
	c)	**Give** at least two differences between ULTs and KLTs.	2
2.	a)	**Differentiate** between deadlock and starvation.	2
	b)	**Explain** the classical synchronization problem concerned with the sharing of reusable resources.	3
	c)	**Elaborate** on mailboxes and their types.	5
3.	a)	**Discuss** why the brute force approach to deadlocks is popular in most desktop operating systems.	1
	b)	**Differentiate** between deadlock prevention and deadlock avoidance.	2
	c)	**Explain** what a checkpoint it and how it helps in recovery from deadlocks.	2
4.	a)	Briefly **comment** on the issues that can arise in multi-user file systems.	3
	b)	**Discuss:** Free block list must be stored on disk rather than main memory.	3
	c)	**Explain** the issues involved in file allocation.	3
		CLO-2: Explore synchronization and resource management issues in operating systems (C3, PLO-2)	
5.	a)	A file contains 3 million records. **Calculate** the average number of disk accesses required to locate a record under each of the following conditions: i. No index is used. (sequential search) ii. A single-level index containing 6,000 entries is used. iii. A two-level index is used, where the higher-level index has 60 entries and the lower-level index has 2,400 entries.	4
	b)	Consider a disk with advertised average seek time of 6 milliseconds, rotational speed of 10,000 rpm, and 1024-byte sectors with 300 sectors per track. Suppose we wish to read a file consisting of 3,600 sectors. **Compare** the times to read the file if: i. It is sequentially allocated. ii. It is randomly allocated.	4
	c)	A disk has 150 tracks (0 – 149). The disk head is initially positioned at track 70. The following disk access requests arrive in order: 25, 38, 142, 130, 10, 95, 45, 8, 120, 100, 3, 55 Using FIFO and SSTF disk scheduling algorithms, schedule the requests and **calculate** the total and average seek length for each.	3
6.		There are three processes P_0, P_1, P_2 and three serially reusable resources A, B, C in the system. The MAX and ALLOCATION matrices are $\begin{bmatrix}1 & 1 & 0\\1 & 1 & 1\\2 & 1 & 1\end{bmatrix}$ and $\begin{bmatrix}0 & 1 & 0\\0 & 0 & 1\\1 & 0 & 0\end{bmatrix}$ respectively. The AVAILABLE vector is $[1 \quad 0 \quad 0]$.	

P.T.O

a) **Determine** if this state is safe. 3

b) P_1 requests one instance of A. **Determine** if this request should be granted. 5

7. There are three processes P_0, P_1, P_2, and three consumable resources R_0, R_1, R_2, in the system. **Determine** if there is a deadlock. Otherwise, **resolve** the system using a series of resource allocation graphs. 3



8. Let us simulate a clinic comprising a nurse, a doctor and a pharmacist. A patient first visits the nurse, who takes his vitals and writes them in a structure VITALS. Then the patient visits the doctor, who reads VITALS and writes in a structure PRESCRIPTION. Finally, the patient goes to the pharmacist who reads PRESCRIPTION and gives medicine to the patient accordingly. The nurse must never write vitals for a new patient until the doctor reads those of the old patient. The doctor must never write a prescription for a new patient until the pharmacist reads that of the old patient. Three patients visit the clinic. Use semaphores to **synchronize**. **Enforce** mutual exclusion wherever required. 8

Nurse	Doctor	Pharmacist

void nurse() {
while(true) {
write_vitals(); } }
	void doctor() {
	while(true) {
	read_vitals();
	write_prescription(); } }
		void pharmacist() {
		while(true) {
		read_prescription();
		give_medicine(); } }

Patient	**Clinic**	

void patient(int i) {
while(true) {
nurse();
doctor();
pharmacist();
get_well();
wait_until_sick_again(); } }
	void main() {
	parbegin(nurse(), doctor(), pharmacist(), patient(1), \\
	patient(2), patient(3)); }	